

Profit and Loss Simulator

Powered by IVolatility

The Position Profit and Loss Simulator, powered by IVolatility.com, allows users to explore the potential profit and loss of hypothetical trades. The user can enter multiple trades, also known as legs, and can customize variables such as stock price, volatility, interest rates, days to expiration and dividends to model theoretical changes. Note that all data is provided with a 20-minute delay. If the markets are closed, the previous trading day's market closing price will be used.

Choose a Stock Symbol, Index Symbol, or Option Symbol

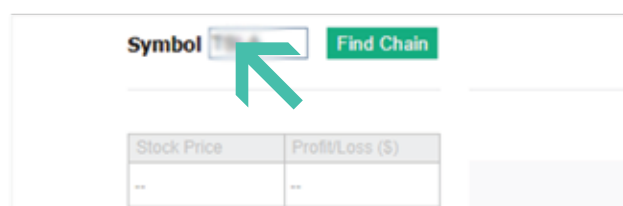
Input any stock, index or option symbol and select 'Find Chain.' Once entered, an options chain pop up window will appear listing all options on the underlying security.

Users can add a position to the PnL simulator from the Options Chains as follows:

- To add the underlying stock or ETF, click the 'Add Stock to Calculator' on the bottom left of the Options Chains window.

Note: users will not be able to add the underlying if the underlying is an index but can instead use a tradeable ETF for an index.

- To add option legs, click on the bid price link to sell, or the ask price link to enter a buy.



Symbol

Stock Price	Profit/Loss (\$)
...	...

The image shows a web browser window displaying the 'Options Chain' for TSLA. The browser's address bar shows the URL 'https://paperkit.net/stock/options-calculator/#view_options_chain?symbol=TSLA&action=Tab'. The page title is 'Options Chain - TSLA - Google Chrome'. The main content area has a header with the text 'Click bid or ask to add short or long option position, click 'add stock' at the bottom to add stock position'. Below this is a table with columns: TSLA, Last, Call, Put, Bid, Ask, Last Tm/Pl, and Volume. The table lists various options with their respective prices and volumes. At the bottom of the page, there is a 'Total Position' section and a button labeled 'Add Stock to Calculator'. A green arrow points to this button. Another green arrow points to the 'Add Stock to Calculator' button at the bottom left of the table.

Position Leg Details

Once all desired legs of the positions are entered users can check the details of each leg and make any corrections. All changes are made in the Position Details window.

Total Position

Price	Bid	Mid	Ask	Delta	Gamma	Theta	Vega
241.20	707.90	709.97	712.05	-399.98	-0.00	4.55	-0.05

Action	Expiry	Strike	Type	Quantity	Price
Buy	Nov 24 2023	20.0	Call	1	220.70
Buy	Nov 24 2023	40.0	Call	1	200.70
Buy	Nov 24 2023	90.0	Call	1	150.75
Buy	Nov 24 2023	140.0	Call	1	135.75

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Total Position

Price	Bid	Mid	Ask	Delta	Gamma	Theta	Vega
241.20	707.90	709.97	712.05	-399.98	-0.00	4.55	-0.05

	Action	Expiry	Strike	Type	Quantity	Price	IV	Delta	Gamma	Theta	Vega
X	Sell	Nov 24 2023	20.0	Call	1	220.70	250.00	-100.00	-0.00	0.30	-0.00
X	Sell	Nov 24 2023	40.0	Call	1	200.70	250.00	-100.00	-0.00	0.59	-0.00
X	Sell	Nov 24 2023	90.0	Call	1	150.75	250.00	-100.00	-0.00	1.40	-0.01
X	Sell	Nov 24 2023	105.0	Call	1	135.75	250.00	-99.98	-0.00	2.27	-0.04

Remove all | Add option leg | Add stock leg

The user can adjust the price of the options/stock manually by typing in the price column. By default, the column shows a 20-minute delayed price from the time the leg was added.

The quantity column allows the user to enter the desired quantity for each leg. The user can also change the expiry, strike, and type of each leg.

Delete any leg by clicking on the (X) icon in the first column.

The other Greek values in the table are calculated and are not available for editing.

The Greeks are represented in shares equivalent. The absolute values of the Greeks are multiplied by the standard contract size and the quantity of options selected in the QTY column.

IMPLIED VOLATILITY OR IV: Implied Volatility is the market's best approximation of future volatility, and it is obtained by plugging the current option price into an option pricing model and backsolving to arrive at the option's implied volatility.

DELTA: The theoretical amount an option's price will change for a corresponding one-point change in the price of the underlying security.

GAMMA: The theoretical change in Delta divided by the dollar change in the price of the underlying security's price. It is the measurement of the rate of change of an option's Delta with respect to the underlying price.

THETA: A measure of how much an option's price theoretically decays with each passing day with the price of the underlying security and implied volatility remaining unchanged.

VEGA: The theoretical measure of change in an option's price in response to a one percentage point change in implied volatility.

Total Position

The Total Position window shows the summed total position Greeks as well as the Bid/Ask and Mid price of all options in the position.

Underlying Price - The user can enter prices and view how those price changes will affect the net Greeks and PnL.

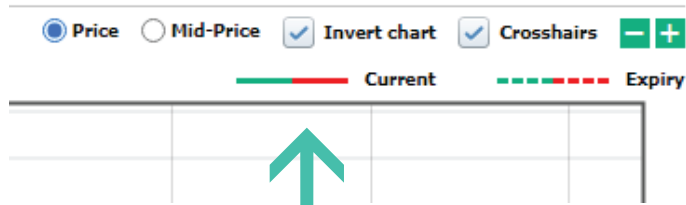
Price - The Bid/Ask prices will be used according to the strategy (Bid for Sell and Ask for Buy).

Mid-price - The mid-price will be used for all position legs.

Invert Chart - The legs will be switched from buy to sell and vice-versa.

Total Position

Price	Bid	Mid	Ask	Delta	Gamma	Theta	Vega
241.20	707.90	709.97	712.05	-399.98	-0.00	4.55	-0.05



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Simulation Panel

To change the volatility assumption the user can manipulate the slider in the first window of the Simulation Panel. The user can also enter a number in the text entry field or drag the slider.

Changes in volatility can be made in "Relative" terms or "Absolute". For example, if the current IV is 20 a 10% relative move would translate to 22%. The 2% is added to the original 20% because the relative move is positive.

To simulate changes in volatility in absolute terms, choose "Absolute" from the drop-down menu. After a 10% absolute change, for example, volatility would equal 30% (10 + 20).

To see how time to expiration affects the position, use the horizontal slider located above the volatility slider. Changes can be made by typing in the number of days in the text-entry field. The user can also drag the horizontal slider with their mouse or click on the calendar icon and choose a date.

Days to expiration always populates with today's date.

Interest Rates

By default, the tool uses SOFR rates in the PnL calculation. However, the user can input their own rates on the first window of the 'Advanced Settings' panel to view how different interest rate assumptions will impact the position.

Dividends

Dividends and dividend yields are provided from the iVolatility database. The user can check and edit them through the "Div/Size" simulation window. The contract size can be edited. The default for the contract size is 100.

The screenshot shows the 'Volatility % Change' section with a slider ranging from -100% to 100%. A dropdown menu is open, showing 'Absolute' selected, with 'Absolute' and 'Relative' as options. A green arrow points to the 'Absolute' option. Below this, the 'Days to Expiry' field is shown with a calendar icon. A green arrow points to the calendar icon. The calendar shows November 2023, with the 24th selected. Below the calendar is a table with columns: Price, Bid, Mid, Ask, Delta, Gamma, Vega. The values are: Price 231.93, Bid 593.70, Mid 599.15, Ask 604.60, Delta -399.98, Gamma -0.00, Vega 6.01, and a final value -0.03.

The screenshot shows the 'Advanced settings' panel with the 'Interest' tab selected. A green arrow points to the 'Interest' tab. The table shows the following data:

Expiration	Interest Rate
11/24/23	5.3193
12/01/23	5.316
12/08/23	5.3202
12/15/23	5.3299
12/22/23	5.3394
12/29/23	5.3451
01/05/24	5.3509
01/19/24	5.3624
02/16/24	5.3769

Buttons for 'Reset' and 'Apply' are at the bottom.

The screenshot shows the 'Advanced settings' panel with the 'Div/Size' tab selected. A green arrow points to the 'Div/Size' tab. The fields are:

- Dividend Amount: 0.00
- Dividend Frequency: Monthly (dropdown)
- Dividend Date (mm/dd/yyyy): 11/22/2023
- Contract Size: 100

Buttons for 'Reset' and 'Apply' are at the bottom.

Stock Price/ PnL Scenario

A table summary of various PnL's across different price points is displayed to the left of the chart. At breakeven the profit is equal to zero. The table defaults to today's date and will change when various inputs are changed (time, price, and expiration). If the user wants to simulate breakeven(s) at expiration, move the days to expiration slider all the way to the right or type "0" in the "Days to Expire" field.

To calculate the PnL for option positions the theoretical option price according to the simulated stock price is used. The Black-Scholes model is used for European-style options and for early exercise of American-style options with an underlying without dividends and for all others Cox-Ross-Rubinstein binomial tree model is used.

Stock Price	Profit/Loss (\$)
60.00	\$59,370.00
90.00	\$57,357.53
120.00	\$48,370.00
150.00	\$36,370.00
180.00	\$24,370.00
210.00	\$12,370.00
240.00	\$370.00
240.93	\$0.00
270.00	-\$11,630.00
300.00	-\$23,630.00

Interactive Chart

Once the strategy is entered into the calculator the line graph will be drawn in the main window automatically. The chart will display the profit and loss as of the day the trade was entered.

The blue horizontal bold line on the graph shows your breakeven point, i.e., where your profit would equal zero.

The user can slide the pointer over the line graph and a crosshair will appear. Use this feature to pinpoint exact profits at a given price level for the underlying. If the user clicks on the chart, it will fix the crosshair and display the profit or loss at that specific underlying price. Click the mouse again to release the crosshair.

Disclaimer

The Profit and Loss Simulator and 20-minute delayed options quotes are provided by IVolatility, and NOT BY OCC. The tool is being provided for educational purposes only. OCC makes no representation as to the timeliness, accuracy or validity of the information and this information should not be construed as a recommendation to purchase or sell a security, or to provide investment advice. For more information or help, please contact us at options@theocc.com.



To zoom in or zoom out on the graph, use the "+" and "-" icons. The user can also drag the graph up or down, left or right, to see any portion of the picture.